2.6 CASE STUDY

Weathering the Storm:

Twitter's Uncertain Future

Twitter, the social network originally based on 140-character text messages, emerged seemingly out of nowhere to take the world by storm. Twitter's basic idea was to marry short text messaging on cellphones with the Web and its ability to create social groups. As the years passed, Twitter expanded beyond simple text messages to article previews, photographs, videos, and animated images, by 2022, it had almost 230 million daily active users worldwide. The 5,000 tweets a day that it began with in 2006 has turned into a deluge of around 10,000 tweets per second and more than 500 million per day worldwide. Special events, such as the Super Bowl and Academy Awards, tend to generate an explosion of tweets. Some celebrities, such as pop star Justin Bieber, have millions of followers (in Bieber's case, around 114 million).

Twitter has a number of important assets, such as user attention, significant audience size (unique visitors), and its searchable database of tweets, which contain real-time audience comments, observations, and opinions. Twitter has become a significant media platform for the distribution of news. However, Twitter has also struggled to develop a profitable business model and display consistent growth, and its financial results and stock price have not matched its popularity and influence on culture and politics. Its user base remains a fraction of the size of Facebook's, TikTok's, and Instagram's.



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In April 2022, Elon Musk, the controversial founder and CEO of electric car company Tesla and space exploration company SpaceX, entered into an agreement to acquire Twitter in a deal that valued the company at $44 billion. After many stops and starts, the purchase finally took place on October 27, 2022, but the drama surrounding Musk has thrown Twitter, its investors, its advertisers, and its users into turmoil, raising serious questions about what the future will bring for Twitter.

Before delving into the current chaos surrounding Twitter, it helps to have some basic background into its business model, revenue model, and evolution. Twitter derives the lion's share of its revenue from advertising, with 90% of its 2021 revenue coming from various advertising products and with more than 95% of them displayed on a mobile device. Twitter offers advertisers a variety of options, including Promoted Ads (formerly called Promoted Tweets), Follower Ads (formerly called Promoted Accounts), and Twitter Takeover (formerly called Promoted Trends). Companies pay to have their tweets and Twitter accounts appear higher in Twitter searches or to reserve a place within Twitter's Trends section, which shows the most-discussed topics of the day. They can also pay to show video advertisements that run prior to video clips and that are embedded within tweets using a product called Twitter Amplify. Video is Twitter's fastest growing advertising segment and has grown to comprise almost two-thirds of Twitter's advertising revenue. All of Twitter's advertising products can be geo-targeted both locally and nationally.

Twitter also continues to refine its data-mining capability, recognizing that its stockpile of customer sentiment about products, services, and marketing efforts is among its most valuable assets. Acquisitions of companies such as Lucky Sort, Topsy Labs, and Gnip have helped Twitter provide better information about user behavior. Twitter licenses its user data to companies seeking to improve their understanding of customer sentiment. The company's data licensing segment accounted for about 11% of its revenues in 2021.

Twitter became a publicly traded company in 2013, with a valuation at that time of about $14 billion. The public offering was viewed as a rousing success, with the stock price jumping almost 75% on its opening day, despite the company's lack of profits. However, its share price declined significantly from its high of more than $74 in December 2013 down to an all-time low of less than $14 in 2016, well below its IPO price. Analysts expressed serious concerns about Twitter's future beyond the obvious lack of profits. Twitter's growth rates were anemic because of unusually poor user retention—60% of new users failed to return to the site the following month. Twitter also had highly unbalanced user engagement, with the top 15% of users accounting for 85% of all tweets. Advertisers also complained that Twitter's advertising products simply were too costly for the sales they generated and that dummy accounts and trolls ran the risk of ruining the Twitter user experience. Acknowledging a need for a change in direction, Chief Executive Officer Dick Costolo stepped down in 2015, replaced by co-founder Jack Dorsey.

Dorsey vowed to narrow the company's focus on its core service, with particular emphasis on breaking news. Twitter moved away from products and features that didn't do enough to enhance the basic user experience. For example, Twitter had hoped that it would become a hub of social e-commerce and rolled out a Buy Now button in 2014 that allowed users to add products to their Amazon shopping carts. However, in 2016,

SOURCES: "Elon Musk Begins Reign of Twitter," by Alexa Corse, *Wall Street Journal*, October 30, 2022; "Attorneys for Musk, Twitter Argue Over Information Exchange," by Associated Press, Usnews.com, September 27, 2022; "Twitter Reports Drop in Revenue, Blames Uncertainty over Elon Musk Deal," by Sarah Needleman, *Wall Street Journal*, July 22, 2022; "Elon Musk and Twitter Will Go to Trial over Their $44 Billion Deal in October," by Kate Conger, *New York Times*, July 19, 2022; "Twitter Sues Elon Musk to Stop Him from Bailing on $44 Billion Deal," by Nick Wingfield, Theinformation.com, July 13, 2022; "Elon Musk Seeks to Abandon $44 Billion Twitter Deal," by Meghan Bobrowsky, *Wall Street Journal*, July 9, 2022; "Elon Musk Got Twitter's Data Dump, Next Comes the Hard Part," by

development on the service halted because of users' lukewarm response, and it was then phased out. Other services that failed to take off, such as Twitter's #Music app, were shelved, and even popular features like its Vine app for short-form video were discontinued. Instead, Twitter made changes to bolster its ability to deliver the news, including increasing the character limit for tweets from 140 characters to 280 characters and showing more promoted news content in user timelines. Twitter redesigned the Explore tab within its app to improve ease of use, added breaking news alerts, and launched dedicated content hubs for significant events that allowed users to follow events the way they might follow a person. It also redesigned its desktop site, aiming to provide a more consistent experience across both its mobile and its web platforms. Twitter also focused on improving its streaming video capability, including livestreaming, highlight video clips, and video-on-demand agreements across a number of verticals including sports, news, gaming, and entertainment.

Developing machine learning capabilities has also been an important initiative for Twitter. It established an internal research group called Cortex that focuses on improving Twitter by enabling advanced artificial technologies. Twitter purchased several artificial intelligence startups, such as Magic Pony, Aiden, and Fabula AI, to bolster this effort. Using AI techniques has enabled Twitter to adjust the way it ranks tweets in users' feeds, emphasizing relevance to individual users instead of simple chronological order. Machine learning algorithms now decide what types of push notifications to deliver to users. Twitter has also eliminated its less effective advertising formats, focusing only on those that deliver the best engagement for advertisers. Twitter is additionally focusing on techniques to promote what it calls conversational health, adjusting its algorithm to ensure that low-quality tweets are less accessible in searches and conversations and changing its API to prohibit links and content from being shared from many accounts at once, which is a common technique used by Twitter bots (dummy accounts that are used to spread misinformation). Although Twitter continues to aggressively identify and remove these accounts, more continue to pop up in their place. Twitter has also tried to improve its handling of habitual abusers and hate groups using its platform by employing machine learning technology to detect network manipulation and online disinformation. Striking a balance between allowing free speech and policing unacceptable hate speech and deliberate misinformation has continued to be a difficult challenge for Twitter.

These changes, in addition to Dorsey's restructuring of Twitter's board and layoffs of 9% of Twitter's workforce, finally jumpstarted Twitter's stagnant growth. In the fourth quarter of 2017, Twitter recorded the first profitable quarter in the company's history, and the company continued that trajectory through the fourth quarter of 2019. Then the Covid-19 pandemic hit, severely impacting Twitter. Although total revenue increased by 7%, it recorded a net loss of $1.14 billion for the year. Adding to the turmoil, an activist investor group that acquired a significant chunk of Twitter's shares sought to have Dorsey removed as chief executive officer. While a truce was reached, leaving Dorsey in place for the time being, his future remained somewhat uncertain. In 2021, Twitter tried to regroup. It introduced several new products, such as Twitter Spaces, a new way to have live audio conversations on Twitter, and Communities, a new way to easily find and connect with people who have similar interests. Twitter also jumped on the creator bandwagon and introduced Ticketed Spaces, Super Follows, and Tips, which give people the ability to pay creators using a variety of payment methods. Twitter also introduced

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Twitter Blue, its first-ever consumer subscription offering. Twitter Blue allows users to pay for exclusive features and perks. Returning to the social e-commerce efforts it had previously abandoned, it also launched a pilot program of a shopping feature called Shop Module that allows businesses to showcase their products at the top of their profiles and allows users to purchase items without having to leave Twitter.

In November 2021, Dorsey resigned. Although Twitter's revenue in 2021 increased from that of 2020, it once again recorded a loss for 2021, this time in part because of the costs of settling a shareholder class action lawsuit for around $810 million. Twitter's stock price declined.

Enter Elon Musk. At the beginning of April 2022, regulatory filings revealed that Musk had personally purchased a 9.2% stake in Twitter, making him its largest shareholder. Musk had previously had a contentious relationship with Twitter, questioning the company's commitment to free speech and wondering if the company was even needed. Musk initially agreed to join Twitter's board but then abruptly reversed course and shortly thereafter offered to buy the rest of the company. Twitter initially moved to block Musk but then reconsidered, and on April 25, accepted his bid to take over the company for $44 billion and go private, giving him sole control.

But not long thereafter, the deal began to unravel. In mid-May 2022, Musk said he would not proceed with the deal unless Twitter could provide him with evidence that less than 5% of the company's user accounts were actually fake. In securities filings, Twitter has long estimated that false or spam accounts represent less than 5% of its total number of active users but has also said that the actual number could be higher than estimated. In early June Musk reiterated his position and accused Twitter of not complying with his request for data on the number of spam and fake accounts. At the end of June, Twitter provided Musk with access to historical tweet data as well as its so-called "fire hose" of tweets: a data dump of the full stream of all tweets made in near real time. Some analysts believed that Musk was using the issue as a way to renegotiate or even scuttle the deal in light of the stock market downturn. On July 8, 2022, Musk announced that he was terminating the deal. In return, Twitter sued Musk in an attempt to make him complete the purchase at the agreed-upon price. In the meantime, Twitter posted an unexpected drop in advertising revenue for the second quarter in 2022, which Twitter and many analysts attributed to Musk's chaotic acquisition attempt. Finally, in October 2022, Musk agreed to proceed with the purchase. However, that is not the end of the story. Twitter remains in turmoil, with its future uncertain.

Case Study Questions

1. What are Twitter's most important assts?
2. How has Twitter's business model evolved since it was founded?
3. Why is machine learning an important initiative for Twitter?
4. What steps did Twitter take to increase revenues in 2021?
5. Why is Twitter's future currently uncertain?

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